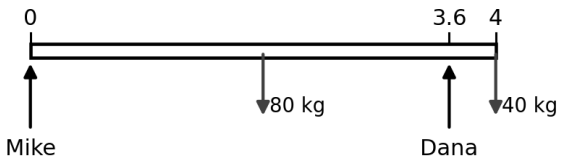
PHYS1408 Practice Exam 3-1

Chapters 12 (Statics) and 13 (Gravity) — 15 multiple-choice questions

$G = 6.67 \times 10^{-11}\ \frac{\text{N m}^2}{\text{kg}^2}$ $\qquad M_{earth} = 5.98 \times 10^{24}\ \text{kg}$ $\qquad R_{earth} = 6.37 \times 10^{6}\ \text{m}$
$g = 9.8\ \text{m/s}^2$

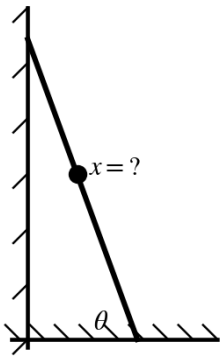
Instructions. Choose the single best answer (a–e). Answers are rounded; pick the nearest choice. For a **maximum** ("how far can he safely…"), pick the largest choice that still works; for a **minimum**, the smallest choice that still works. A calculator and your 3×5 notecard are allowed.

1. A 80 kg sofa 4 m long is carried by Mike and Dana. Mike has one end; Dana holds the sofa 0.4 m from the other end. A helper sets a 40 kg box of books on the very end of the sofa that Dana is holding. How much force must Dana exert now?



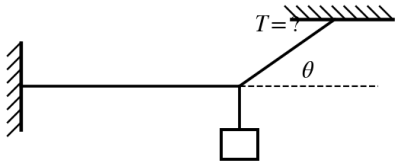
a) 305 N
b) 436 N
c) 588 N
d) 871 N
e) 1,180 N

2. A 120 kg man (with the load he carries) climbs a uniform 6 m, 25 kg ladder that leans against a frictionless wall at 70° to the floor. The coefficient of static friction between the ladder and the floor is 0.3. How far up the ladder can he safely climb?



a) 5.0 m
b) 5.5 m
c) 6.0 m
d) 6.5 m
e) 7.0 m

3. A 15 kg light hangs from a knot. One cord runs from the knot to the ceiling, making a 35° angle with the ceiling. A second cord is exactly horizontal and is anchored in the wall. What is the tension in the cord connected to the ceiling?



a) 84.3 N
b) 147 N
c) 179 N
d) 210 N
e) 256 N

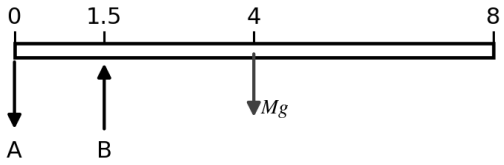
4. A block of jello with $Y = 25{,}000$ N/m^2 is 2 m wide, 3 m long and 0.15 m high. A 400 kg mass resting on top compresses it. By how much does the height shrink?

a) 0.400 mm
b) 3.92 mm
c) 26.1 mm
d) 52.3 mm
e) 78.4 mm

5. A 300 kg load hangs from a circular wire 3 m long with a diameter of 0.0015 m and a Young's modulus of 2×10^{11} N/m^2. By how much does the wire stretch?

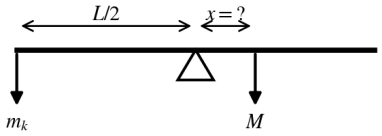
a) 0.00255 m
b) 0.00624 m
c) 0.00832 m
d) 0.0196 m
e) 0.0250 m

6. A uniform diving board is 8 m long and has a mass of 600 kg. It rests on two supports: support A at the back end (x = 0) and support B 1.5 m from that end. Nothing is standing on the board. What is the magnitude of the force from support B?



a) 5,880 N
b) 9,800 N
c) 15,700 N
d) 21,600 N
e) 31,400 N

7. A seesaw 5 m long pivots at its center. A 20 kg kid sits at one end. How far from the center must a 80 kg adult sit on the other side so that the seesaw balances? The seesaw itself has a mass of 40 kg.



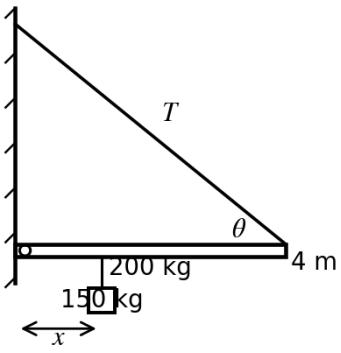
a) 0.417 m
b) 0.625 m
c) 1.25 m
d) 1.88 m
e) 3.12 m

9. What is the acceleration due to gravity on a 90 kg astronaut at a height of 500 km above the Earth's surface?

a) 8.45 m/s^2
b) 9.83 m/s^2
c) 1,600 m/s^2
d) 2,240 m/s^2
e) 3,140 m/s^2

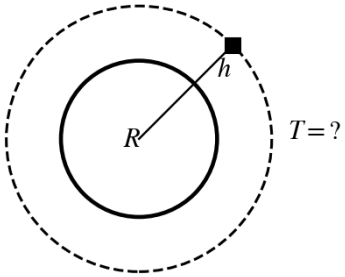
8. A uniform 4 m, 200 kg beam sticks straight out from a wall on a hinge. A cable runs from the far end of the beam back to the wall above the hinge, making a 40° angle with the beam. The cable can hold at most $T_{max} = 3{,}000$ N. A 150 kg mass hangs from the beam a distance x from the wall. What is the largest x the cable can support?

a) 0.658 m
b) 0.921 m
c) 1.29 m
d) 2.58 m
e) 3.59 m



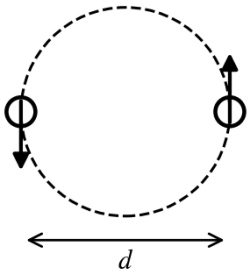
10. What is the period of a satellite in a circular orbit around the Earth at an altitude of 20,000 km above the Earth's surface?

a) 1.40 hours
b) 1.88 hours
c) 5.92 hours
d) 7.82 hours
e) 11.8 hours



11. Two equal-mass stars (5×10^{30} kg each) orbit their common center of mass, halfway between them. Their separation is $d = 2 \times 10^{9}$ m. What is the period of their orbit?

a) 3.02 hours
b) 4.27 hours
c) 6.04 hours
d) 8.55 hours
e) 17.1 hours



12. A rocket fired straight up from the Earth's surface rises to a maximum height of 600 km. At what speed was it launched? (Ignore air resistance.)

a) 1.66 km/s
b) 2.32 km/s
c) 3.28 km/s
d) 7.56 km/s
e) 11.2 km/s

13. The Earth and the moon (mass 8×10^{22} kg) are 380,000 km apart, center to center. How far from the center of the Earth toward the moon, as a percentage of the Earth–moon distance, is the point where the gravitational pull toward the Earth equals the pull toward the moon?

a) 10.4%
b) 50.0%
c) 80.8%
d) 89.6%
e) 98.7%

14. What is the escape speed from the surface of the Moon (mass 7.4×10^{22} kg, radius 1.74×10^{6} m)?

a) 1.68 km/s
b) 2.38 km/s
c) 4.76 km/s
d) 6.66 km/s
e) 9.32 km/s

15. A satellite is to circle the Earth once every 8 hours. At what altitude above the Earth's surface must it orbit?

a) 13,900 km
b) 20,300 km
c) 62,800 km
d) 91,500 km
e) 1.28×10^{5} km